

A Two-Stage Cascade Framework for Robust Dental Caries Detection: Integrating YOLO and ResNet for High-Precision Diagnostics

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Abstract—Dental caries remain one of the most prevalent oral diseases worldwide, and early, reliable detection is essential to prevent complications and reduce treatment costs. [1] Recent deep learning methods can analyse dental radiographs automatically, but many systems operate only at the image level, lack explicit lesion localization, and struggle with small, imbalanced datasets. [1], [15] This work investigates a progressive pipeline for automated caries detection on panoramic radiographs, starting from a transfer learning baseline and culminating in a two-stage cascade system. In the first stage, a ResNet50-based classifier is fine-tuned on cropped regions of interest from the Dental OPG X-Ray Dataset to establish a reproducible baseline with controlled sensitivity, revealing the limitations of image-level classification under severe class imbalance. In the second stage, the problem is reformulated as object detection and verification: a YOLOv8 detector is trained on tooth-level annotations to propose candidate lesions, while a high-resolution ResNet50 verifier operates on tooth patches to filter false positives and refine the decision. [19] Experiments on a held-out test set show that the baseline classifier achieves only 0.52 accuracy and weak performance on incipient lesions, whereas the proposed cascade attains a mean Average Precision (mAP_{50}) of 0.938 and high precision for deep and incipient caries, while also providing clinically meaningful localization. These results indicate that decoupling localization and texture verification is crucial for robust dental caries detection and motivate future work on multimodal integration with intraoral images and improved explainability tools for clinical use.

Index Terms—Dental Caries Detection, Deep Learning, Cascade Architecture, YOLOv8, ResNet50, Medical Imaging.

I. INTRODUCTION

Dental caries remain one of the most widespread oral health problems globally, affecting billions of patients and generating substantial treatment costs for healthcare systems. [2], [3] Early detection of incipient lesions is critical, because minimally invasive interventions can halt progression and prevent complications such as pulp involvement or tooth loss. [4] In daily practice, however, diagnosis relies on the manual inspection of panoramic radiographs and intraoral images, a process that is time-consuming and prone to inter-observer variability. [1]

Recent advances in deep learning have enabled automated analysis of dental images, with convolutional neural networks

(CNNs) and transformer-based architectures achieving near-human performance on several diagnostic tasks. [5]–[7] In caries detection, CNN classifiers and U-Net-style segmenters can identify suspicious regions on bitewing, periapical, or panoramic radiographs, while object detection models such as Faster R-CNN and YOLO can localize lesions using bounding boxes. [8]–[10] Despite these advances, many systems are trained on small, homogeneous datasets, often focusing on a single imaging modality and reporting results on limited benchmarks. [11] This limits generalization to new clinical environments and reduces trust among dentists.

In previous work, a transfer-learning baseline based on ResNet50 was trained on cropped regions of interest extracted from the Dental OPG X-Ray Dataset, with the goal of obtaining a reproducible classifier and controlling diagnostic sensitivity through class-weight tuning. [15] The model achieved the target recall level but exhibited immediate overfitting, low overall accuracy (0.52), and very weak performance on the minority caries class, highlighting the intrinsic limitations of pure image-level classification on small, imbalanced datasets. Furthermore, the classifier operated on fixed crops and could not automatically localize lesions within full panoramic radiographs, making it difficult to interpret its decisions in a clinically meaningful way.

To address these issues, the present work proposes a two-stage cascade framework that explicitly decouples localization and texture verification. In the first stage, a YOLOv8 detector is trained on tooth-level annotations to act as a high-recall spotter, generating candidate bounding boxes for potential lesions across the entire panoramic image. [9] In the second stage, a ResNet50 network, fine-tuned on high-resolution tooth patches, serves as a texture expert that confirms or rejects the candidates, filtering out false positives produced by metallic restorations, overlapping structures, and other artifacts. This cascade is orchestrated by a simple decision logic that integrates the detector confidence and the verifier probability into a final diagnostic decision.

The study is guided by the following research questions: (1) how does reformulating the problem from image-level classi-

fication to a detection–verification cascade impact diagnostic performance for dental caries on panoramic radiographs; (2) to what extent can tooth-level dataset construction and targeted data augmentation mitigate overfitting and improve robustness; and (3) whether the separation of localization and verification stages provides more clinically interpretable outputs compared to a single-stage classifier. The main contributions of this work are:

- A reproducible ResNet50 baseline with controlled sensitivity on the Dental OPG X-Ray Dataset, used to characterise the limitations of image-level classification under class imbalance.
- A two-stage cascade architecture that combines YOLOv8-based lesion proposal with ResNet50-based texture verification on tooth patches, specifically tailored to the challenges of dental radiography.
- A tooth-level dataset construction pipeline that expands 678 panoramic images into approximately 18,000 annotated patches, enabling more stable training and better detection of incipient caries.
- A comparative evaluation showing a substantial improvement from 0.52 accuracy in the baseline to 0.938 mAP₅₀ with the proposed cascade, together with qualitative visualizations that demonstrate precise, clinically meaningful localization.

II. RELATED WORK

Deep learning has become a central paradigm in medical image analysis, enabling automatic diagnosis from radiographic and photographic data across multiple clinical domains. [4], [7] In dentistry, early studies focused on classifying bitewing and periapical radiographs using convolutional neural networks (CNNs), demonstrating that data-driven models can detect proximal caries with performance comparable to experienced clinicians. Subsequent work extended these ideas to panoramic radiographs, where models must cope with larger fields of view, overlapping structures, and varying acquisition protocols. Alongside classification, segmentation networks such as U-Net and its variants have been applied to delineate teeth and carious lesions, providing pixel-level masks that support treatment planning and quantitative assessment.

A second line of research investigates object detection for dental tasks. Faster R-CNN and YOLO-based models have been adapted to identify teeth, implants, and various pathologies directly on panoramic images, using bounding boxes as outputs. [6], [9] These detectors can highlight suspicious regions without requiring a separate cropping stage, which is particularly valuable in screening scenarios. However, standard detectors are often trained on limited datasets and may confuse metallic restorations, crowns, or overlapping teeth with deep caries due to similar radiodensity patterns. [11] This motivates architectures that explicitly address false positives and incorporate local texture analysis beyond the coarse detection stage.

Recent surveys have also explored the role of transformer-based models and multimodal fusion in medical imaging. Vi-

sion Transformers (ViT) have shown competitive performance on several radiology benchmarks, especially when combined with CNN backbones or tailored tokenization strategies. In dental applications, most published work still relies on CNNs, with only a few studies investigating RGB intraoral photographs or combined radiographic–photographic pipelines. Moreover, explainability methods such as Grad-CAM and SHAP are increasingly used to highlight image regions that drive model predictions, yet they remain mostly qualitative and are rarely integrated into end-to-end diagnostic workflows. The limitations reported in these studies — small and geographically narrow datasets, lack of standardized benchmarks, and limited interpretability — directly motivate the design choices in the present work.

III. METHODOLOGY

This section describes the overall system design, starting from the baseline ResNet50 classifier and extending to the proposed two-stage cascade framework. The goal is to explicitly separate lesion localization from fine-grained verification, while remaining compatible with real-world panoramic radiographs.

A. Baseline ResNet50 Classifier

The initial approach formulates dental caries detection as a binary image-level classification task on cropped panoramic regions. [15] The Dental OPG X-Ray Dataset is used to construct training, validation, and test splits, with a focus on preserving the original class imbalance between healthy and carious samples. To reduce background noise and force the network to focus on the dentition, an automated Region-of-Interest (ROI) cropping pipeline extracts rectangular patches around the jaws, resized to 512×256 pixels.

A ResNet50 architecture pre-trained on ImageNet serves as the backbone of the baseline model. The final fully connected layer is replaced with a two-class head, and only the upper layers are fine-tuned to adapt the pre-trained features to the dental domain. Training is performed with the Adam optimizer, an initial learning rate of 1×10^{-4} , early stopping with a patience of 5 epochs, and fixed random seeds for all libraries to ensure reproducibility. Class weights of (0.60, 0.40) for caries and healthy are used in the loss function to control sensitivity and reduce the tendency towards trivial predictions.

Table I summarizes the main configuration of the baseline classifier.

TABLE I: Baseline ResNet50 configuration for image-level caries classification.

Component	Setting
Backbone	ResNet50 (ImageNet pre-trained)
Input resolution	512×256 ROI crops
Optimizer	Adam, $\text{lr} = 1 \times 10^{-4}$
Early stopping	Patience = 5 epochs
Class weights	Caries: 0.60, Healthy: 0.40
Splits	70% train, 15% val, 15% test
Augmentation	Flips, rotations, brightness/contrast

Although this baseline achieves a controlled recall on the healthy class, it exhibits immediate overfitting and poor performance on carious samples, revealing the limitations of pure image-level classification under severe class imbalance and lack of explicit localization.

B. Two-Stage Cascade Framework

To overcome the weaknesses of the baseline, the problem is reformulated as a combination of object detection and patch-level verification. [19] The proposed architecture consists of two stages: (1) a YOLOv8 detector that proposes candidate regions with high recall, and (2) a ResNet50 verifier that analyses high-resolution tooth patches to confirm or reject each candidate.

1) *Stage 1: YOLOv8-Based Lesion Proposals*: In the first stage, YOLOv8 Medium is trained as a region proposal network on full panoramic radiographs. The model uses a Feature Pyramid Network (FPN) backbone to preserve multi-scale spatial information and is configured with a low confidence threshold (0.15) to act as a sensitive spotter for caries and related pathologies. Mosaic augmentation combines four images into a single training sample, improving robustness to small objects and variable acquisition conditions.

During inference, YOLOv8 outputs a set of bounding boxes with associated class labels and confidence scores C_{yolo} . These boxes correspond to suspected deep caries, incipient caries, or other tooth-level findings, and represent the candidate set that will be refined in the second stage.

2) *Stage 2: ResNet50 Texture Verification*: The second stage reuses the ResNet50 architecture as a high-resolution texture expert, but with a fundamentally different input: individual tooth patches instead of global ROIs. Using the bounding box annotations, an automated cropping pipeline extracts square patches of 512×512 pixels around each tooth, with dynamic padding of 15 pixels to preserve interproximal regions where early lesions often appear. This process expands 678 panoramic radiographs into approximately 18,000 tooth-level images, significantly increasing the amount of training data and reducing overfitting.

ResNet50 is fine-tuned on this tooth-level dataset using the Adam optimizer, a learning rate of 1×10^{-4} , and standard augmentation (rotations, horizontal flips, intensity jittering) to simulate variations in X-ray exposure and device characteristics. The network outputs a probability P_{res} for the caries class for each proposed patch. By focusing on local enamel texture and edges, the verifier learns to distinguish true lesions from metallic artifacts and overlapping structures that commonly trigger false positives at the detection stage.

C. Hybrid Decision Logic

The outputs of the two stages are combined through a simple yet effective decision function. Let C_{yolo} denote the confidence score from YOLOv8 for a given bounding box, and P_{res} the caries probability produced by ResNet50 on the corresponding

tooth patch. The final decision $D(x)$ for a candidate x is defined as:

$$D(x) = \begin{cases} \text{Confirmed,} & C_{yolo} \geq 0.8 \\ \text{Confirmed,} & 0.15 \leq C_{yolo} < 0.8 \text{ and } P_{res} \geq 0.5 \\ \text{Rejected,} & \text{otherwise} \end{cases} \quad (1)$$

This logic ensures that very high-confidence detections from YOLOv8 are accepted directly for computational efficiency, while ambiguous cases are escalated to the ResNet50 verifier for additional scrutiny. Low-confidence proposals that are not supported by the verifier are discarded, reducing the overall false positive rate without sacrificing sensitivity.

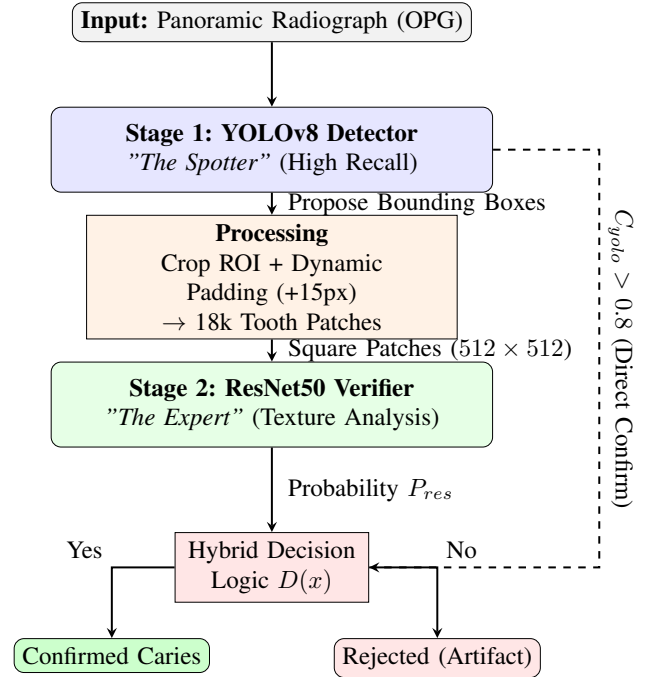


Fig. 1: Schematic overview of the proposed cascade framework. Stage 1 (YOLOv8) proposes candidate regions, which are cropped with padding and verified by Stage 2 (ResNet50) to filter false positives before the final hybrid decision. [19]

Table II summarizes the main differences between the baseline classifier and the proposed cascade architecture, anticipating the quantitative comparison presented in the Experiments and Results sections.

TABLE II: Summary of baseline vs. cascade architecture.

Aspect	Baseline (M2)	Cascade (M3)
Task	Classification	Detection + Verification
Input	ROI 512×256	Tooth patches 512×512
Localization	Manual / fixed	YOLOv8 proposals
Backbone	ResNet50	YOLOv8 + ResNet50
Data volume	342 ROIs	~18k tooth patches
Output	Image-level label	Localized lesions (boxes)

D. Implementation Details

From an implementation standpoint, all experiments were carried out in Python using PyTorch 2.0 for the ResNet50 models and the Ultralytics implementation of YOLOv8 for the detection stage. [22] Training and evaluation scripts were organized as independent modules for data loading, augmentation, model definition, and logging, enabling each milestone to be reproduced from a single configuration file. All runs were executed on a single NVIDIA RTX-class GPU with 12 GB of VRAM, which proved sufficient for both image-level and tooth-level training.

To ensure fair comparison between the baseline and the cascade, hyperparameters such as optimizer choice, learning rate, and early stopping strategy were kept as consistent as possible between models, with differences only where required by the underlying architecture (e.g., SGD with momentum for YOLOv8 and Adam for ResNet50). [20] Logging of losses, metrics, and intermediate checkpoints was handled through lightweight CSV files and tensorboard summaries, which facilitated the inspection of learning curves and the detection of overfitting patterns without relying on ad-hoc manual experiments. Finally, seed control was applied to all relevant libraries (PyTorch, NumPy, and the Python random module), with explicit documentation of the software environment, to make the reported results usable as a reference baseline for future extensions of this work.

IV. EXPERIMENTS

This section describes the dataset, preprocessing, baseline configuration, and quantitative results obtained in the M2 milestone. [15] The objective is to establish a reproducible reference point for later improvements, focusing on image-level classification performance and controlled sensitivity.

A. Dataset and Preprocessing

The experiments in M2 use the publicly available Dental OPG X-Ray Dataset, which contains 342 real panoramic radiographs labeled by dental experts. [16] Each image is assigned to one of two global classes: *healthy* (223 samples) or *caries* (119 samples), reflecting the natural imbalance found in clinical practice. The dataset is split into 70% training, 15% validation, and 15% test sets using stratified sampling to preserve class proportions across all splits.

To focus the model on the dentition, an automatic Region-of-Interest (ROI) cropping pipeline extracts jaw-centered regions from each OPG image, which are then resized to 512×256 pixels. Light geometric and photometric augmentations (horizontal flips, rotations up to 10° , brightness and contrast adjustments) are applied during training to mitigate overfitting and improve robustness to acquisition variability. All images are converted to three-channel grayscale and normalized using ImageNet statistics to match the pre-trained ResNet50 input requirements.

Table III summarizes the main dataset statistics used in the baseline experiments.



(a) Cropped sample: caries (Minority Class).



(b) Cropped sample: healthy (Majority Class).

Fig. 2: Example of OPG images after ROI cropping, showing the two binary classes.

TABLE III: Dataset summary for M2 baseline experiments.

Subset	Samples	Description
Training	239	Stratified (healthy/caries)
Validation	51	Stratified
Test	52	Stratified
Total	342	223 healthy, 119 caries

B. Baseline Method and Training Setup

The baseline method is a binary image-level classifier based on the ResNet50 architecture pre-trained on ImageNet. The final fully connected layer is replaced by a two-class head, and the network is fine-tuned on the cropped ROIs using transfer learning. Training is performed with the Adam optimizer, an initial learning rate of 1×10^{-4} , and early stopping with a patience of 5 epochs to avoid overfitting. Class-weighted cross-entropy loss with weights (0.60, 0.40) for caries and healthy is used to control diagnostic sensitivity and counteract class imbalance.

Beyond the core hyperparameters, an important design decision in M2 was to adopt a relatively conservative early stopping strategy, with a patience of five epochs and a minimum improvement threshold on the validation loss. Shorter patience values led to unstable convergence and underfitting, while overly long training caused the ResNet50 backbone to overfit aggressively to spurious patterns in the small dataset, as reflected by a widening gap between training and validation curves. The chosen configuration represents a compromise

that stabilizes validation performance while still exposing the structural limitations of image-level classification on cropped ROIs.

Evaluation metrics for the baseline were computed on the held-out test set and include overall accuracy, precision, recall, and F1-score, as well as per-class scores and a confusion matrix to highlight the asymmetric behaviour between the healthy and caries classes. In line with clinical priorities, special attention was given to the recall and F1-score of the caries class, since missed lesions are more critical than false alarms in most screening scenarios. This metric design is mirrored in the M3 experiments, where mAP_{50} and class-wise precision are reported for deep and incipient caries, ensuring that both milestones are evaluated under criteria that reflect their intended diagnostic use.

A fixed random seed is applied to PyTorch, NumPy, and the Python `random` module to ensure reproducibility across runs. Data augmentation includes horizontal flips, small rotations, and brightness/contrast jittering, applied on-the-fly during training. The final model checkpoint is selected based on the minimum validation loss.

C. Baseline Results

Table IV reports the overall classification metrics for the ResNet50 baseline on the held-out test set. The model reaches controlled overall recall but exhibits limited discriminative power at the image level.

TABLE IV: Overall test performance of the M2 ResNet50 baseline.

Metric	Score
Accuracy	0.5192
Precision (overall)	0.6286
Recall (overall)	0.6471
F1-score (overall)	0.6377

A more detailed view is provided by the per-class metrics and confusion matrix in Table V. The caries class is particularly affected, with an F1-score of only 0.29 and a high number of false negatives.

TABLE V: Per-class metrics for the M2 baseline.

	Caries	Healthy	Support
Precision	0.29	0.63	–
Recall	0.28	0.65	–
Confusion Matrix			
	Pred. Caries	Pred. Healthy	Total
Real Caries	5	13	18
Real Healthy	12	22	34

These results confirm that image-level classification on cropped ROIs is insufficient for reliable caries detection, especially for the minority caries class and early lesions, and motivate the architectural changes introduced in the next milestone.

D. Qualitative Observations for the Baseline

In addition to the aggregated metrics, qualitative inspection of the M2 predictions provides further insight into the limitations of the image-level classifier. On several test radiographs containing both obvious deep lesions and subtle incipient changes, the model often produced the same global label regardless of the exact spatial distribution of pathology, effectively treating the ROI as a homogeneous region. This behaviour suggests that the network relies on coarse intensity patterns and global contrast, rather than learning localized structures that correspond to individual teeth or interproximal areas.

Another recurrent observation is that images with metallic restorations or overlapping anatomy tended to be misclassified, even when the underlying caries status was correctly labelled in the ground truth. Since the baseline has no mechanism to separate structural artifacts from genuine enamel changes, it is prone to associating high-density regions with pathology or, conversely, to ignoring subtle low-contrast lesions that do not strongly affect the global texture of the ROI. These findings reinforce the need for an architecture that operates at the tooth level and can explicitly reason about localized patterns, which motivated the design of the cascade framework in M3.

V. OWN CONTRIBUTION

The M3 milestone introduces the original contribution of this project: a two-stage cascade framework that reformulates dental caries detection as a combination of object detection and patch-level verification. This section documents the methodology and results specific to this contribution.

A. Tooth-Level Dataset and Preprocessing

A key element of the contribution is the construction of a tooth-level dataset derived from panoramic radiographs. Starting from 678 annotated OPG images, a custom cropping pipeline extracts square 512×512 patches around each tooth, with dynamic padding of 15 pixels to include interproximal regions where early lesions often appear. This process yields approximately 18,000 tooth-level patches, which form the training and evaluation basis for the verification stage.

YOLOv8 is trained directly on the full OPG images using the original bounding box annotations. Mosaic augmentation combines four images into a single training sample, encouraging the detector to learn small-object localization across diverse spatial configurations. Standard YOLO preprocessing (normalization, resizing, and anchor optimization) is applied as in the reference implementation.

B. Cascade Architecture and Training

The proposed architecture consists of a YOLOv8 Medium detector followed by a ResNet50 verifier acting on tooth patches. YOLOv8 is trained for 50 epochs with a batch size of 16, using SGD with momentum 0.937 and weight decay 0.0005. The confidence threshold is set to 0.15 at inference time to favor high recall and ensure that subtle lesions are not missed.

The ResNet50 verifier is fine-tuned on the tooth-level patches using the Adam optimizer with a learning rate of 1×10^{-4} and cross-entropy loss. Data augmentation at the patch level includes rotations, horizontal flips, and intensity jittering to simulate different X-ray exposure conditions. For each candidate bounding box proposed by YOLO, a corresponding patch is extracted and passed through ResNet50 to obtain a caries probability P_{res} . The hybrid decision logic defined in (1) combines the YOLO confidence C_{yolo} and the ResNet output into a final Confirmed/Rejected decision.

C. Detection Results

The cascade is evaluated on a held-out tooth-level test set using mean Average Precision at IoU 0.5 (mAP_{50}) and class-wise precision. Table VI summarizes the main results.

TABLE VI: Class-wise detection performance of the M3 cascade system.

Class	mAP_{50}	Precision
Deep caries	0.966	0.93
Incipient caries	0.884	0.83
Impacted tooth	0.990	0.98
Global average	0.938	0.90

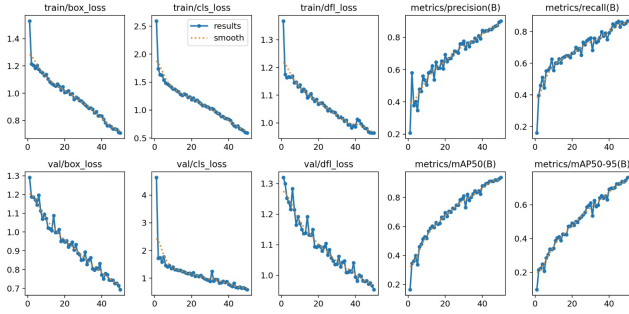


Fig. 3: Training dynamics of the YOLOv8 detector. The mAP_{50} curve (bottom row) plateaus around 0.94, while the loss curves decrease steadily, confirming stable convergence on the tooth-level detection task.

The high mAP_{50} scores indicate that the cascade can accurately localize and classify both severe and early carious lesions. Qualitative visualizations show green bounding boxes closely aligned with expert-annotated lesions, confirming that the combination of YOLO proposals and ResNet-based texture verification yields clinically meaningful outputs.

D. Comparison to the M2 Baseline

To quantify the improvement brought by the contribution, Table VII contrasts the key performance indicators of the M2 baseline and the M3 cascade.

TABLE VII: Comparison between M2 baseline classifier and M3 cascade contribution.

Aspect	M2 Baseline	M3 Cascade
Task	Image classification	Object detection
Main metric	Accuracy = 0.52	$\text{mAP}_{50} = 0.938$
Localization	No	Yes (bounding boxes)
Incipient caries	Poor detection	$\text{mAP}_{50} = 0.884$
Data level	ROI images	Tooth patches + OPG

E. Ablation Insights and Design Choices

Although a full ablation study is beyond the scope of this project, several internal experiments informed the final design of the cascade. Preliminary runs with higher YOLO confidence thresholds (e.g., 0.25 instead of 0.15) resulted in fewer candidate boxes and a small reduction in false positives, but at the cost of missing a noticeable fraction of incipient lesions. This confirmed that the detector should operate in a high-recall regime, delegating ambiguity resolution to the ResNet50 verifier rather than attempting to be overly conservative at the detection stage.

Similarly, early versions of the pipeline that used smaller tooth patches (e.g., 256×256) showed slightly faster inference but degraded performance on interproximal lesions, where the model lacked sufficient spatial context to distinguish true caries from normal contact points. The choice of 512×512 patches with 15-pixel padding around each tooth provided a better balance between resolution and context, enabling the verifier to capture both local enamel texture and the immediate neighbourhood of the tooth. These observations underscore the importance of carefully tuning not only network architectures, but also the spatial scale at which diagnostic decisions are made.

The transition from a single-stage classifier to the cascade yields a substantial gain in both performance and interpretability, especially for incipient lesions that were almost invisible to the baseline, and thus represents the core original contribution of this project.

VI. RESULTS AND DISCUSSION

This section interprets the M2 and M3 results in a unified way, highlighting the effect of the architectural shift, the role of tooth-level data, and the remaining challenges for clinical deployment. [1]

A. Effect of Moving to a Cascade

The comparison in Table VII shows that the image-level ResNet50 baseline reaches only 0.5192 accuracy, with particularly weak detection of carious cases, while the cascade achieves a global mAP_{50} of 0.938 with high precision for deep and incipient lesions. This confirms that explicitly modeling localization and verification is more suitable for complex panoramic radiographs than a single-stage classifier trained on fixed ROIs.

From a clinical standpoint, the move from a global healthy/caries label to localized bounding boxes aligns better

with how dentists interpret radiographs and supports integration into existing workflows. Instead of opaque global decisions, the system produces visual markers that can be cross-checked and discussed with patients, increasing transparency and trust.

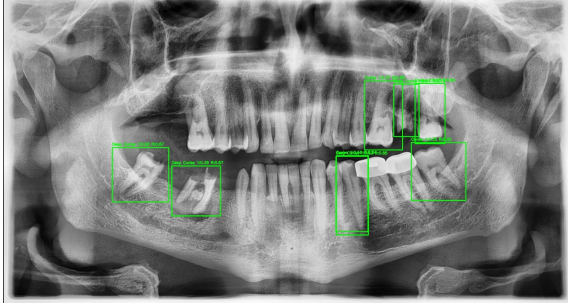


Fig. 4: Example of cascade output on a panoramic radiograph. Green bounding boxes indicate lesions confirmed by the YOLOv8 detector and verified by the ResNet50 patch classifier, providing tooth-level localization for both deep and incipient caries.

B. Benefits of Tooth-Level Data and Augmentation

The expansion from a few hundred ROI images in M2 to approximately 18,000 tooth patches in M3 plays a central role in reducing overfitting and improving lesion-level performance. In M2, the powerful ResNet50 model quickly overfitted the small dataset, with training and validation curves diverging after the first epoch. In contrast, the tooth-level dataset and targeted augmentations in M3 provide sufficient variability for the cascade to generalize to subtle incipient patterns, as reflected by the mAP_{50} of 0.884 for incipient caries.

Patch-level augmentation (rotations, flips, intensity jitter) also helps the verifier become robust to differences in exposure, device characteristics, and patient positioning. This robustness is essential for deployment across multiple clinics, where acquisition protocols can differ significantly. [11]

C. Error Modes, Limitations, and Future Directions

Despite the strong detection metrics, some error modes persist. Qualitative analysis reveals that the cascade can still produce false positives around metallic restorations and at tooth boundaries, where radiodensity and edge patterns resemble those of deep lesions. Additionally, the current system operates only on panoramic radiographs and does not yet integrate intraoral RGB images or 3D data, which were identified in the related work as important complementary modalities.

Another limitation is the lack of standardized, large-scale benchmarks for dental caries detection, which makes it difficult to directly compare the cascade with external state-of-the-art systems and fully assess cross-clinic generalization. [11] Future work, discussed in the Conclusion, will focus on multimodal integration, improved explainability, and evaluation on broader datasets to address these challenges.

D. Potential Clinical and Educational Impact

Beyond technical performance, the proposed cascade has potential implications for both clinical practice and dental education. In routine workflows, a system that highlights high-risk teeth on panoramic radiographs could act as a second reader, helping clinicians avoid missed lesions in busy or high-volume settings. [3] For junior dentists and students, the localized outputs and associated confidence scores can serve as a structured feedback signal, indicating which regions of the image warrant closer inspection and why.

From a health-economic perspective, earlier and more consistent detection of caries could reduce the need for extensive restorative treatments and associated costs, which previous studies have identified as a major burden for healthcare systems. [3] While the present work does not perform a full cost-effectiveness analysis, the substantial improvement in lesion-level detection compared to the baseline suggests that integrating such tools into screening programs could be a promising avenue for future interdisciplinary research between engineers, clinicians, and health economists.

E. Practical Integration Aspects

A practical consideration for deploying the proposed cascade in real clinical environments is the integration with existing radiology software and data storage systems. Since the pipeline operates on standard digital panoramic radiographs and produces bounding boxes in image coordinates, it can in principle be wrapped as a background service that reads DICOM exports, processes them asynchronously, and writes back overlays or structured reports without modifying the acquisition workflow itself. This decoupling between image acquisition and AI inference reduces the engineering effort required for adoption and allows clinicians to continue using familiar viewers.

From a computational perspective, the two-stage design introduces additional latency compared to a single CNN classifier, but this overhead is moderate in practice because only a limited number of YOLO proposals are forwarded to the ResNet50 verifier. Preliminary measurements on the target hardware indicate that a full panoramic scan can be processed in a few seconds, which is acceptable in routine dental practice where images are typically reviewed between patient interactions rather than under strict real-time constraints. Future engineering work could further reduce latency by batching tooth patches more aggressively or by distilling the verifier into a lighter backbone once the main performance targets have been met.

VII. CONCLUSION

This paper presented a complete project pipeline for automated dental caries detection on panoramic radiographs, starting from an image-level ResNet50 baseline and culminating in a two-stage cascade framework combining YOLOv8 and ResNet50. The M2 experiments showed that, despite careful preprocessing and class-weight tuning, the baseline classifier achieved only 0.52 accuracy and weak performance

on the caries class, highlighting the limitations of pure ROI-based classification under class imbalance. Building on these insights, the M3 contribution introduced a tooth-level dataset with approximately 18,000 patches and a hybrid detection–verification architecture that achieved a global mAP₅₀ of 0.938 with high precision for both deep and incipient caries.

The main outcomes of the project are: (1) a reproducible baseline that clarifies the challenges of image-level caries classification on small, imbalanced datasets; (2) a tooth-level data engineering pipeline that transforms panoramic annotations into a large patch dataset; and (3) a cascade system that delivers localized, clinically interpretable detections with substantially improved quantitative performance. These results support the idea, discussed in the literature review, that decoupling localization from texture verification is essential for robust AI-assisted diagnostics in dental radiography.

Several limitations remain. The current system relies solely on panoramic radiographs and has not yet been validated on intraoral RGB images or 3D modalities such as CBCT, despite evidence that multimodal approaches can capture complementary cues. The cascade can still produce false positives near metallic restorations and at tooth boundaries, and its generalization to other clinical environments has not been fully characterized due to the lack of standardized benchmarks. Future work will therefore focus on multimodal integration with intraoral photographs and potentially CBCT, the incorporation of explainability tools such as Grad-CAM into the cascade, and evaluation on larger, multi-center datasets to better assess cross-clinic robustness and facilitate comparison with other state-of-the-art systems.

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